[METRIC]
A-A-52514
August 14, 1995
SUPERSEDING
MIL-P-62423(AT)
14 August 1991
(see 5.5)

COMMERCIAL ITEM DESCRIPTION

POWDER, POLYESTER-ALUMINUM (METRIC)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers a two-component powder containing polyester and alloyed aluminum particles mechanically distributed in a uniform mixture (40 polyester, 60 Si-Al). This powder is primarily intended for use as an abradable coating for use in environments up to 343 degrees Celsius (°C).

2. SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein, the materials used shall be in accordance with the manufacturer's materials specification. The use of recovered material made in compliance with regulatory requirements is acceptable providing all requirements of this CID are met (see 5.5).

2.1.1 Formulation. The two component mixture shall consist of the percentage by weight as shown in table I.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E, Warren, MI 48397-5000.

FSC 9630

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:11Z
Check the source to verify that this is the current version before use.

TABLE I. Component percentages by weight.

Component	Percentage (%)
A (Aluminum alloy) B (Polyester)	59 - 61 Remainder

2.1.2 Particle size. The particle size distribution by weight shall be as shown in table II.

TABLE II. Particle size distribution by weight.

Component	Sieve designation 1/		% by weight	Weight passed through sieve	Weight retained on sieve
	Micrometer	Mesh			
A(Aluminum alloy)	150 75 45	100 200 325	99 minimum (min) 85 min 70 min	X X X	
B (Polyester)	150 90 45	100 170 325	99 min 85 min 25 maximum	X X X	

1/ Sieve designation in accordance with ASTM E11.

2.1.3 Chemical composition. The chemical composition of components A and B shall conform to the percent by weight specified in the following:

a. Component A (silicon-aluminum). The chemical composition of component A shall be as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:11Z
Check the source to verify that this is the current version before use.

TABLE III. Chemical composition of component A.

Elements	% by weight	
	Minimum	Maximum
Silicon	11.0	13.0
Iron	----	0.8
Copper	----	0.3
Other elements,each	----	0.2
Other elements,total	----	0.6
Aluminum	Remainder	

b. Component B. Component B shall be a completely aromatic polyester system.

2.1.4. Properties. The material shall meet the following properties requirements when applied (plasma sprayed) to iron, and titanium base alloys.

a. Bond strength. The bond strength of the plasma-sprayed material shall be not less than 7.6 megapascals (MPa).

b. Cup test. The amount of draw before spalling of the coating shall not be less than 1.52 millimeter (mm).

2.2 Unit containers. Unless otherwise specified (see 5.3), the unit containers shall be capable of holding the following quantities of powder: 0.5, 1.0, 2.5, 5.0, and 10.0 kilograms (kg). The containers shall be sealed moisture proof units and large enough to permit mixing the powder within before use. The containers shall be capable of protecting the powder against damage or contamination until it is ready to be used.

2.3 Identification markings. The identification markings shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code, part number, and lot number and the part or identification number (PIN) (see 5.3).

2.3.1 Labeling. Unit containers and packages shall be conspicuously labeled "CAUTION: Containers contain powder aluminum. Use approved respirator."

2.3.2 Container. Each container shall be marked "Polyester-Aluminum Powder".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:11Z
Check the source to verify that this is the current version before use.

3. QUALITY ASSURANCE PROVISIONS

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (see 5.2).

5. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

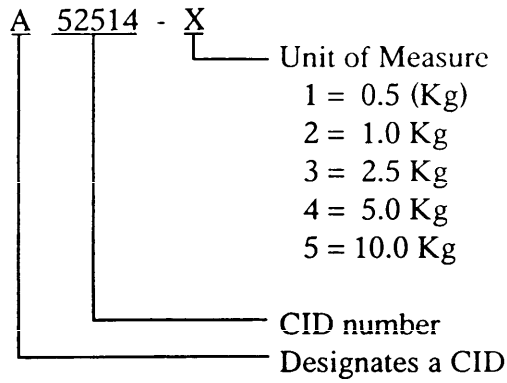
5.1.1 Non-Government publications. Copies of ASTM E11 "Standard Specification for Wire Cloth Sieves for Testing Purposes" are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

5.2. Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Selection of applicable level and packaging requirements.
d. Unit container and PIN number.
e. Unit container size if other than specified.

5.3 Part or identification number (PIN). The PINs to be used for powder acquired to this CID, are created as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:11Z
Check the source to verify that this is the current version before use.



5.4 Cross reference data. Powder conforming to this CID is interchangeable/substitutable with powder conforming to MIL-P-62423(AT). Cross reference to former part number is as indicated in table IV.

TABLE IV. Cross reference.

Unit Container size	PIN	Former PIN No.
0.5 Kg	A52514-1	M62423- 0.5Kg
1.0 Kg	A52514-2	M62423- 1.0Kg
2.5 Kg	A52514-3	M62423- 2.5Kg
5.0 Kg	A52514-4	M62423- 5.0Kg
10.0 Kg	A52514-5	M62423-10.0Kg

5.5 Supersession data. This CID supersedes Textron Lycoming (Allied Signal) internal/ specification M3955E, dated 26 April 1982.

5.6 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

PREPARING ACTIVITY:
Army - AT

(Project 9630-0030)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:11Z
Check the source to verify that this is the current version before use.